

05_Assignment_Yuliia_Chernysheva

September 28, 2026

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September 27, 2026
CSC 2267

1 CIS 2267 - Topic 5 - Obtaining Data

In this assignment, you will investigate sources of data, and consider the legal and ethical restrictions on data use.

1.1 Data Usage

Increasingly, ours is an information society. Information (data) is a competitive advantage. Information is power.

Ethical and Legal Restrictions on Using Data: Just because you can doesn't mean you should. - Data is private property, commonly known as Intellectual Property (IP) - In the US, the creator of the data owns a copyright whether or not she registers it with the government - Copyright law varies among countries - Generally in USA, the default is to allow usage; in much of the rest of the world, especially the European Union (EU), the default is to prohibit. There are wide variations, so check your local jurisdiction. - Using data created by others requires their permission - Using data created by others requires proper attribution and citation - Data has proprietary, confidentiality and competitive-advantage aspects. To make it exceedingly obvious, the UK calls its legislation "Trade-Secrets" act. - Especially military documents and data can be classified as confidential, secret, top secret. You can do serious jailtime for violations. - Your employer should have appropriate policies and procedures for using internally generated and external data - Governments generally prohibit the use of personally-identifiable data, and have strict rules for creating personally identifiable data - Likewise research institutions, such as universities, have "use of humans in research" policies and procedures - Non-commercial usage, fair use, and creative commons regulations generally provide royalty-free use of data in teaching at universities where there is no profit component. - Government-generated data is usually freely used for most purposes, as the data is in the public domain, being taxpayer funded. For example, US Census Bureau and Bureau of Labor Statistics data. - Commercial use of external data usually involves royalties, user fees and subscriptions, with contractual terms and conditions for using, storing, modifying, combining and reselling resulting data products. - State sponsored and private attackers may try to steal your IP. How will you guard against it? - **LEGAL REQUIREMENTS FOR DATA USE AND STORAGE VARY SUBSTANTIALLY AMONG COUNTRIES. YOU MUST COMPLY WITH ALL LAWS AND REGULATIONS WHEREVER YOUR EMPLOYER HAS OPERATIONS AND DATA USAGE.**

1.2 Data Usage and Data Quality Examples

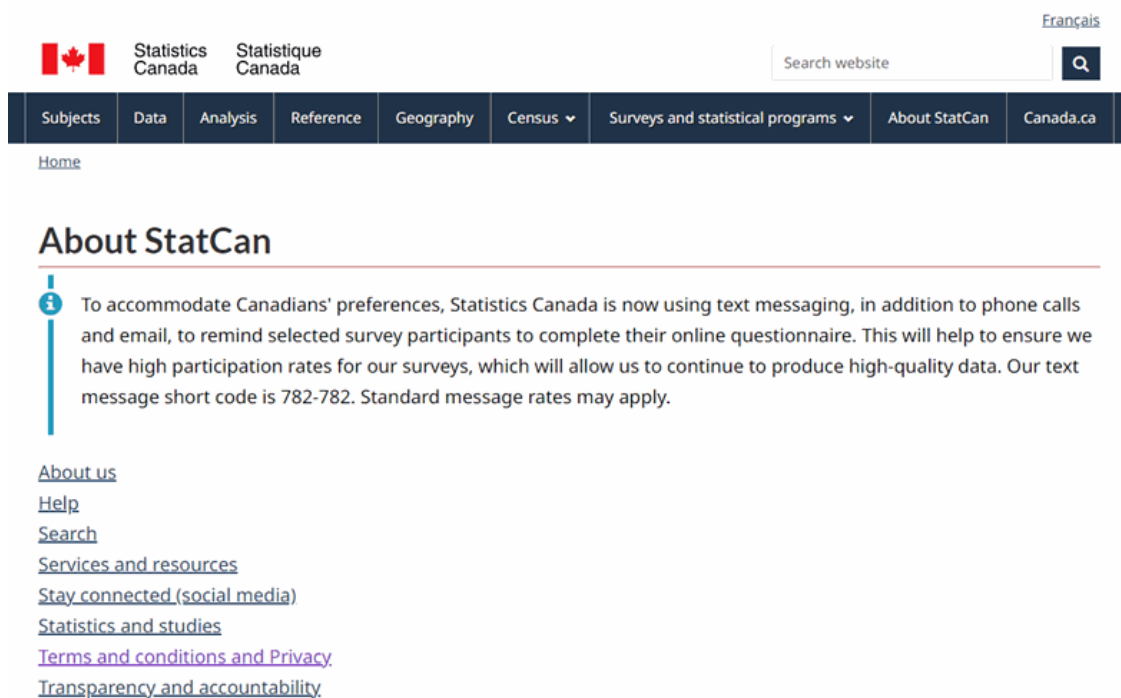
Please open and read the links below to expand your knowledge so that you can become an ethical data analyst, and a more valuable employee ready for promotions and pay raises.

(NOTE: All screenshots and their link locations below are taken from *Statistics Canada* website, the English language home page of which is: <https://www.statcan.gc.ca/en/start> all links accessed and screenshots taken on April, 2022.)

To illustrate data quality and permissioned use of data, *Statistics Canada* was selected because: - Canada is right across the border - It is a more manageable easy-to-use website (35 million people in Canada vs. 350 million in USA) - All Canadian governmental data is centralized in this one agency instead of spread across hundreds of government agencies like USA

1.2.1 1. What is Statistics Canada (StatCan)? What are its capabilities?

<https://www.statcan.gc.ca/en/about/statcan?MM=1>



Please scroll through all the links to see everything StatCan does. Explore links of particular interest to you or relevance to this course.

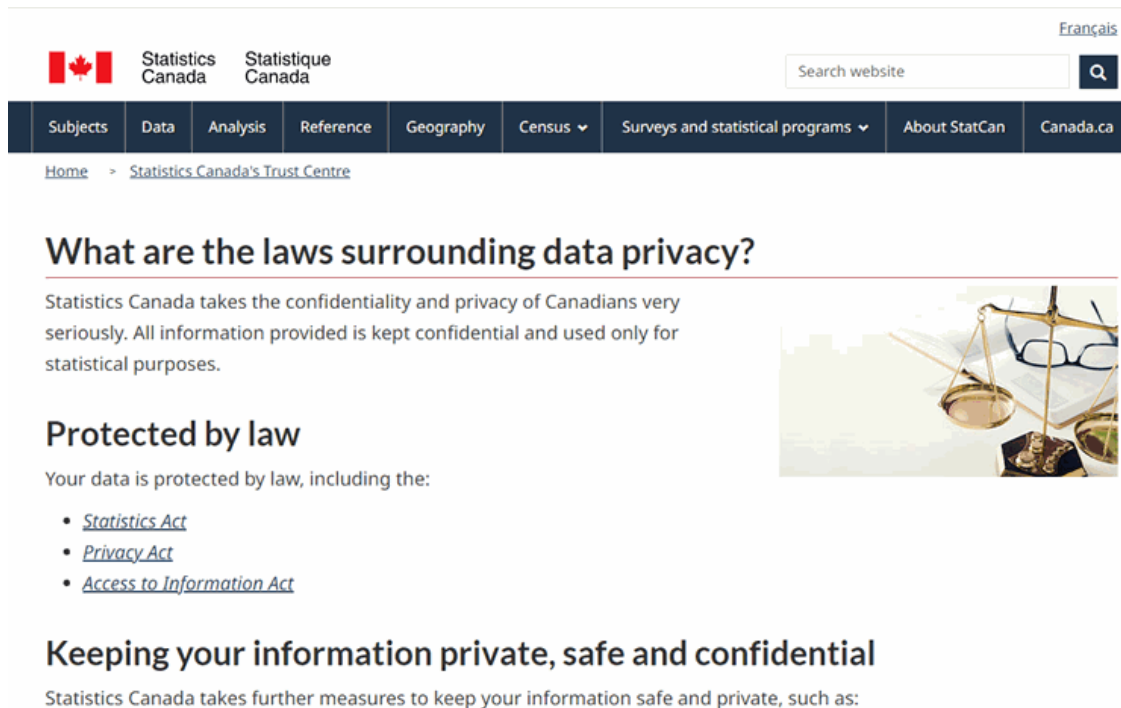
Write a brief summary of your findings:

Summary: Statistics Canada is Canada's central national statistical agency operating under the Statistics Act. Its core capabilities include conducting nationwide censuses, collecting comprehensive economic, social, and demographic data, maintaining centralized public databases, and providing reliable official statistics for research, policy-making, and commercial use.

1.2.2 2. What are Canadian laws surrounding data privacy?

Remember, you must comply with national and local laws in every country/locality where you have activities.

<https://www.statcan.gc.ca/en/trust/laws?MM=as>



The screenshot shows the Statistics Canada website. At the top, there is a navigation bar with the Canadian flag, the text 'Statistics Canada' and 'Statistique Canada', a search bar, and a 'Français' link. Below this is a dark blue navigation menu with links for 'Subjects', 'Data', 'Analysis', 'Reference', 'Geography', 'Census', 'Surveys and statistical programs', 'About StatCan', and 'Canada.ca'. The main content area has a breadcrumb trail: 'Home > Statistics Canada's Trust Centre'. The title 'What are the laws surrounding data privacy?' is prominently displayed. Below the title, a paragraph states: 'Statistics Canada takes the confidentiality and privacy of Canadians very seriously. All information provided is kept confidential and used only for statistical purposes.' To the right of this text is an image of a pair of glasses and a balance scale. Below the paragraph, the heading 'Protected by law' is followed by the text 'Your data is protected by law, including the:' and a bulleted list of three acts: 'Statistics Act', 'Privacy Act', and 'Access to Information Act'. Further down, the heading 'Keeping your information private, safe and confidential' is followed by the text 'Statistics Canada takes further measures to keep your information safe and private, such as:'.

Please scroll through all the links to see everything StatCan does. Explore links of particular interest to you or relevance to this course.

Summary: Canadian data privacy and confidentiality are strictly protected under federal legislation, primarily The Statistics Act, The Privacy Act, and The Access to Information Act. These laws mandate that data collected from individuals and businesses must be kept strictly confidential, backed by absolute legal protections that prohibit sharing data with external entities like police or courts. Furthermore, the framework ensures rigorous safety through strict anonymization of outputs, explicit consent requirements, advanced technological security, and full employee accountability.

1.2.3 3. What are the ownership and usage requirements for content provided by StatCan?

Please scroll through all the links to see everything StatCan does. Explore links of particular interest to you or relevance to this course.

<https://www.statcan.gc.ca/en/reference/copyright?MM=as>

Ownership and usage of content provided on this site

Materials on this website were produced and/or compiled by Statistics Canada for the purpose of providing Canadians with access to information about the programs and services offered by the Government of Canada.

The material on this site is covered by the provisions of the [Copyright Act](#), by Canadian laws, policies, regulations, and international agreements.

The Statistics Canada Open Licence governs the use of most data products and other materials that are published by Statistics Canada. This licence allows you to use Statistics Canada information without restrictions on sharing and redistribution, for commercial and non-commercial purposes.

- [Statistics Canada Open Licence](#)
- [Statistics Canada Open Licence Agreement - Frequently asked questions \(FAQ\)](#)
- [Third-party materials](#)
- [Trademark notice](#)
- [Guidelines for linking to the Statistics Canada website](#)

Summary: StatCan retains intellectual property rights over its information, but makes its content widely accessible under specific legal frameworks. Users are granted the right to use, analyze, and distribute data and derivative products, provided they comply with the agency’s licensing terms, avoid misrepresenting the data, and give proper source attribution.

1.2.4 4. What is StatCan Open License Agreement for use of its content?

This is most important. It details the terms and conditions for using content. This is StatCan’s licensing agreement. Every organization will have such an agreement, but with different provisions.

<https://www.statcan.gc.ca/en/reference/licence>

Statistics Canada Open Licence

This licence is issued on behalf of Her Majesty the Queen in Right of Canada, as represented by the Minister for Statistics Canada ("Statistics Canada") to you (an individual or a legal entity that you are authorized to represent).

The following are terms governing your use of the Information. Your use of any Information indicates your understanding and acceptance of the terms below. If you do not agree to these terms, you may not use the Information.

Statistics Canada may modify this licence at any time, and such modifications shall be effective immediately upon posting of the modified licence on the Statistics Canada website. Your use of the Information will be governed by the terms of the licence in force as of the date and time you accessed the Information.

Definitions

"Information" means the compilation of non-confidential results from any Statistics Canada activities, including data files, databases, public use microdata files, publications, tables, graphs, maps, reports and text for which Statistics Canada is the owner or a licensee of all intellectual property rights and made available to you in accordance with this licence, at cost or no cost, either on the Statistics Canada website or by other means as a result of a contract for goods or services.

Please read the entire agreement as representative of all licensing agreements.

Especially important to us is the worldwide, royalty-free, non-exclusive license:

Licence Grant Subject to this licence, Statistics Canada grants you a worldwide, royalty-free, non-exclusive licence to: - use, reproduce, publish, freely distribute, or sell the Information; - use, reproduce, publish, freely distribute, or sell Value-added Products; and, -sublicence any or all such rights, under terms consistent with this licence.

Summary: The StatCan Open Licence is a worldwide, royalty-free, non-exclusive agreement that allows users to freely use, reproduce, publish, distribute, or sell both raw information and value-added products (such as dashboards, applications, or models), as long as standard terms and attribution rules are respected.

1.2.5 5. StatCan's Open License FAQs

<https://www.statcan.gc.ca/en/reference/licence-faq>

Statistics Canada Open Licence - Frequently Asked Questions (FAQ)

1. [What is an "open" licence?](#)
2. [What information sharing rights come with the Statistics Canada Open Licence?](#)
3. [Does the Statistics Canada Open Licence apply to all Statistics Canada products?](#)
4. [If I already have a Statistics Canada Licence Agreement for a public use microdata file, do I need to contact you for termination?](#)
5. [Is personal information provided to Statistics Canada protected?](#)
6. [Does the Statistics Canada Open Licence apply to third party materials found on Statistics Canada's website?](#)
7. [Do I need to sign the Statistics Canada Open Licence?](#)
8. [Do I need to provide sourcing for every table or graph I create using Statistics Canada data?](#)
9. [Am I permitted to sell Statistics Canada data "as is" or value-added products that I have created?](#)
10. [Will Statistics Canada issue a written permission if I request it?](#)
11. [Am I permitted to provide a link to your website?](#)
12. [Am I permitted to include a screen shot of your website?](#)

Please scroll through all the links to see everything StatCan does with respect to its Open Licence. Explore links of particular interest to you or relevance to this course.

Summary: The Open Licence FAQ clarifies that commercial monetization and selling value-added products are fully permitted without prior registration or signature. It also specifies that users must provide proper source attribution, while allowing truncated citations (e.g., "Statistics Canada, [year]") when generating bulk outputs like multiple plots in Python.

1.2.6 6. Have we complied with the license agreement regarding posting links and screenshots into this document?

Read the FAQ section, and decide if this document is in compliance. Support your decision.

Summary: Yes, this document complies fully. In accordance with FAQ guidelines, all links are text-based URLs that open directly without framing or using official government logos/symbols as navigational buttons. Furthermore, screenshots do not isolate official logos, and each section includes proper source attribution and direct hyperlinks.

1.2.7 7. How Do We Properly Cite StatCan Content in Our own Derivative Products (i.e. your data analytics)?

<https://www150.statcan.gc.ca/n1/pub/12-591-x/12-591-x2021001-eng.htm>



Statistics

Canada

Statistique

Canada

Search website

Q

Subjects

Data

Analysis

Reference

Geography

Census

Surveys and statistical programs

About StatCan

Canada.ca

Home

>

12-591-X

How to Cite Statistics Canada Products

Release date: January 11, 2021

More information

PDF version

On this page

Introduction

This guide has been developed for authors, editors, researchers, academics, students, librarians and data librarians. It provides examples of how to build bibliographic references when citing Statistics Canada products and materials, citation style suggestions by subject matter and contact information for further questions and inquiries. Universities, educational institutions, corporations and other users may have their own guidelines for citing resources. This guide does not intend to displace those requirements, providing guidance on the best method when referencing Statistics Canada's products rather than acting as a "how-to" and "must use" guide.

Notice: Information from Statistics Canada must be acknowledged whenever it is used.

Authors and researchers must give full credit for any Statistics Canada data, analysis and other content material used or

Please read the entire page.

This is an example. Every organization will have its own citation specifications. Often the exact citation is provided to copy and paste into your work.

Summary: Proper citation requires integrating recognized academic styles (such as APA, MLA, or Chicago) into analytical reports or code documentation. Citations must include the author (Statistics Canada), release year, full title with table or catalog numbers, medium format specifiers (e.g., [Data table]), and a direct URL or DOI.

Example (APA Style):

Statistics Canada. (2019). Table 13-10-0096-05 Life satisfaction, satisfied or very satisfied, by age group [Data table]. <https://www150.statcan.gc.ca/t1/tbl1/en/tv.action?pid=1310009605>

1.2.8 8. Example of StatCan provided citation to copy and paste into your work based on this data.

<https://www150.statcan.gc.ca/t1/tbl1/en/tv.action?pid=2510002201>

Type of electricity generation ¹	2013	2014	2015	2016	2017
	Kilowatts				
Total installed capacity ²	130,922,302 ^r	135,088,014	141,396,648	144,375,482	145,213,838
Hydraulic turbine ³	76,520,368 ^r	78,570,149	79,213,900	80,448,736	80,763,622
Wind power turbine ⁴	6,716,982 ^r	8,495,912	11,128,205	11,974,153	12,402,715
Tidal power turbine ⁵	20,000	20,000	20,000	20,000	20,000
Solar power ⁶	171,112 ^r	264,659	1,874,745	2,085,908	2,297,863
Total thermal	47,493,840 ^r	47,737,294	49,159,798	49,846,685	49,729,638
Conventional steam turbine ⁷	20,589,158 ^r	21,068,218	21,293,778	21,469,335	21,295,779
Nuclear steam turbine ⁸	14,033,000 ^r	14,033,000	14,033,000	14,033,000	14,033,000
Combustion turbine ⁹	11,949,596 ^r	11,692,141	12,724,685	13,167,708	13,205,566
Internal combustion turbine ¹⁰	922,086 ^r	943,935	1,108,335	1,176,642	1,195,293

Symbol legend:

^r revised

How to cite: Statistics Canada. [Table 25-10-0022-01 Installed plants, annual generating capacity by type of electricity generation](#)
DOI: <https://doi.org/10.25318/2510002201-eng>

Notice the “How to Cite” at bottom of screenshot.

Official StatCan Citation:

Statistics Canada. Table 25-10-0022-01 Installed plants, annual generating capacity by type of electricity generation.

DOI: <https://doi.org/10.25318/2510002201-eng>.

1.2.9 9. How Does StatCan Ensure Data Quality?

Remember the saying: “garbage in; garbage out”.

<https://www23.statcan.gc.ca/imdb/p2SV.pl?Function=getSurvey&Id=1227927>

Annual Electric Power Generating Stations Survey (EPGS)

Annual Electric Power Generating Stations Survey (EPGS)	Variable(s)	Summary of changes	Other reference periods	Related products
Detailed information for 2018				
Status:	Active			
Frequency:	Annual			
Record number:	2193			
The purpose of this survey is to obtain information on the supply of, and/or demand for, energy in Canada.				
Data release - November 13, 2020				
• Questionnaire(s) and reporting guide(s) • Description • Data sources and methodology • Data accuracy				
Description				
The purpose of this survey is to obtain information on the supply of, and/or demand for, energy in Canada. This information serves as an important indicator of Canadian economic performance, and is used by all levels of government in establishing informed policies in the energy area. The private sector also uses this information in the corporate decision-making process.				

Please read the entire page to appreciate how StatCan ensures data quality.

Summary: To avoid “garbage in, garbage out,” StatCan utilizes a rigorous multi-stage quality framework. This includes standardized questionnaires via the Integrated Business Statistics Program (IBSP), multi-source frame cross-referencing, automated collection and processing error edits, direct respondent follow-ups, strict outlier isolation, and pre-release macro-coherence validation.

1.2.10 10. Three Examples of StatCan Data Products

A. Infographics <https://www150.statcan.gc.ca/n1/en/type/analysis?MM=1>

2. [Statistics Canada - Infographics](#)

Journals and periodicals: 11-627-M

Description: Every year, Statistics Canada collects data from hundreds of surveys. As the amount of data gathered increases, Statistics Canada has introduced infographics to help people, business owners, academics, and management at all levels, understand key information derived from the data. Infographics can be used to quickly communicate a message, to simplify the presentation of large amounts of data, to see data patterns and relationships, and to monitor changes in variables over time.

These infographics will provide a quick overview of Statistics Canada survey data.

[Less](#)

Release date: 2022-04-08

B. StatCan Gender, Diversity and Inclusion In the United States, we call this “Diversity, Equity and Inclusion”, often abbreviated DEI.

https://www.statcan.gc.ca/en/topics-start/gender_diversity_and_inclusion

The screenshot shows the Statistics Canada website interface. At the top, there is a navigation bar with the Canadian flag, the text "Statistics Canada" and "Statistique Canada", a search bar, and a language selector for "Français". Below this is a horizontal menu with links: "Subjects", "Data", "Analysis", "Reference", "Geography", "Census", "Surveys and statistical programs", "About StatCan", and "Canada.ca". The main content area features a large blue banner with the text "GENDER, DIVERSITY AND INCLUSION" and "STATISTICS" next to a circular logo. Below the banner, there is a section titled "Find data on" with four buttons: "Accessibility statistics", "Dimensions of Poverty Hub", "Immigration and ethnocultural diversity statistics", and "Language statistics". To the right of this section, there are social media icons for Facebook, Instagram, LinkedIn, YouTube, and Twitter, along with a prompt to "Sign up to My StatCan to get updates in real-time." Below the social media links, there is a section titled "Disaggregated Data Action Plan" with a brief description of the plan and a small logo.

Explore links of particular interest to you or relevance to this course.

C. Example of a research study based on StatCan data, with references and notes

This is an example of what your end-user clients (internal or external) need your data analysis for. You, yourself, might be doing both the data analytics and the research report. Glad you're taking this course to learn how?

<https://www150.statcan.gc.ca/n1/pub/36-28-0001/2022003/article/00006-eng.htm>

The screenshot shows the Statistics Canada website interface. At the top, there is a navigation bar with the Canadian flag, the text 'Statistics Canada / Statistique Canada', a search bar, and a 'Français' link. Below this is a menu with categories: Subjects, Data, Analysis, Reference, Geography, Census, Surveys and statistical programs, About StatCan, and Canada.ca. The breadcrumb trail reads: Home > 36-28-0001 > Main page. The main heading is 'Economic and Social Reports' followed by the title 'A profile of children with affirmative responses to the 2016 census questions on difficulties with activities of daily living'. The authors are listed as Thomas Charters, Christoph Schimmele, and Rubab Arim. The release date is March 23, 2022. A DOI link is provided: https://doi.org/10.25318/36280001202200300006-eng. There are links for 'More information' and 'PDF version'. A section titled 'On this page' is visible. Below this is the 'Abstract' section, which begins with: 'This study presented a sociodemographic profile of children aged 0 to 14 years with affirmative responses by parents to the questions on the 2016 Census long-form questionnaire about difficulties with activities of daily living. The filter questions, derived from the Disability Screening Questions, were designed to identify persons who are likely to have a disability for an adult population aged 15 years and older. Therefore, relatively little is known about their suitability for a'.

Please read the report, noting its use of data and overall structure. Notice the format of the citations in the “References” section.

Summary: Statistics Canada provides diverse data products, including Infographics that visually translate complex socioeconomic trends into accessible insights, specialized hubs like Gender, Diversity and Inclusion offering disaggregated data and intersectional tools for equity analysis, and comprehensive Research Studies (such as peer-reviewed analytical reports) that demonstrate how official microdata is structured, analyzed, and professionally cited.

1.3 Your Turn

Having researched Statistics Canada’s policies and offerings, find a dataset of interest to you. Load it into a Python `DataFrame` and perform some brief analyses, including a chart or graph. Conclude your analysis with a properly formed citation.

```
[1]: import pandas as pd
from urllib import request
from zipfile import ZipFile
import seaborn as sns
```

```
import matplotlib.pyplot as plt
```

```
[2]: # Load the zip file to disk
zip_url = 'https://www150.statcan.gc.ca/n1/tbl/csv/11100125-eng.zip'
request.urlretrieve(zip_url, filename='11100125-eng.zip')

# Extract the files and list their names
file_names = list()
with ZipFile('11100125-eng.zip',mode='r') as zip:
    zip.extractall()
    for file in zip.infolist():
        file_names.append(file.filename)
    print(file.filename, file.compress_size, file.file_size)
data_file_name = [f for f in file_names if 'metadata' not in f.lower()][0]
```

```
11100125.csv 651456 6130873
11100125_MetaData.csv 4903 16537
```

```
[3]: # Read it into a Python DataFrame
food_spending = pd.read_csv(data_file_name)
food_spending.head()
```

```
[3]:
```

	REF_DATE	GEO	DGUID	Statistic	\
0	2010	Canada	2016A000011124	Average expenditure per household	
1	2010	Canada	2016A000011124	Average expenditure per household	
2	2010	Canada	2016A000011124	Average expenditure per household	
3	2010	Canada	2016A000011124	Average expenditure per household	
4	2010	Canada	2016A000011124	Average expenditure per household	

	Food expenditures, summary-level categories	UOM	UOM_ID	SCALAR_FACTOR	\
0	Food expenditures	Dollars	81	units	
1	Food purchased from stores	Dollars	81	units	
2	Bakery products	Dollars	81	units	
3	Bread and unsweetened rolls and buns	Dollars	81	units	
4	Bread	Dollars	81	units	

	SCALAR_ID	VECTOR	COORDINATE	VALUE	STATUS	SYMBOL	TERMINATED	\
0	0	v54531258	1.1.1	7850.0	NaN	NaN	NaN	
1	0	v54531259	1.1.2	5709.0	NaN	NaN	NaN	
2	0	v54531260	1.1.3	581.0	NaN	NaN	NaN	
3	0	v54531261	1.1.4	273.0	NaN	NaN	NaN	
4	0	v64481413	1.1.180	207.0	NaN	NaN	NaN	

	DECIMALS
0	0
1	0
2	0

```
3      0
4      0
```

```
[4]: # --- Exploratory Data Analysis (EDA) ---
print("=" * 50)
print(f"=> Dataset Dimensions (Rows, Columns): {food_spending.shape}")
print("=" * 50)

print("\n=> General Information on Data Types and Memory:")
food_spending.info()

print("\n" + "=" * 50)
print("=> Statistical Summary (Describe):")
print("=" * 50)
display(food_spending.describe())
```

```
=====
=> Dataset Dimensions (Rows, Columns): (36634, 16)
=====

=> General Information on Data Types and Memory:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 36634 entries, 0 to 36633
Data columns (total 16 columns):
#   Column                                                                 Non-Null Count  Dtype
---  -
0   REF_DATE                                                                36634 non-null  int64
1   GEO                                                                    36634 non-null  object
2   DGUID                                                                   36634 non-null  object
3   Statistic                                                                36634 non-null  object
4   Food expenditures, summary-level categories  36634 non-null  object
5   UOM                                                                    36634 non-null  object
6   UOM_ID                                                                36634 non-null  int64
7   SCALAR_FACTOR                                                         36634 non-null  object
8   SCALAR_ID                                                            36634 non-null  int64
9   VECTOR                                                                36634 non-null  object
10  COORDINATE                                                            36634 non-null  object
11  VALUE                                                                31833 non-null  float64
12  STATUS                                                                4801 non-null   object
13  SYMBOL                                                                0 non-null      float64
14  TERMINATED                                                            0 non-null      float64
15  DECIMALS                                                             36634 non-null  int64
dtypes: float64(3), int64(4), object(9)
memory usage: 4.5+ MB

=====
=> Statistical Summary (Describe):
```

```
=====
```

	REF_DATE	UOM_ID	SCALAR_ID	VALUE	SYMBOL	TERMINATED	\
count	36634.000000	36634.0	36634.0	31833.000000	0.0	0.0	
mean	2015.550035	81.0	0.0	200.625829	NaN	NaN	
std	3.965100	0.0	0.0	790.075481	NaN	NaN	
min	2010.000000	81.0	0.0	1.000000	NaN	NaN	
25%	2012.000000	81.0	0.0	18.000000	NaN	NaN	
50%	2015.000000	81.0	0.0	45.000000	NaN	NaN	
75%	2019.000000	81.0	0.0	114.000000	NaN	NaN	
max	2023.000000	81.0	0.0	13999.000000	NaN	NaN	

	DECIMALS
count	36634.0
mean	0.0
std	0.0
min	0.0
25%	0.0
50%	0.0
75%	0.0
max	0.0

```
[5]: # Filter data for Canada and convert VALUE column to numeric type (coerce
      ↪ errors to NaN)
df_plot = food_spending[food_spending['GEO'] == 'Canada'].copy()
df_plot['VALUE'] = pd.to_numeric(df_plot['VALUE'], errors='coerce')

# Select specific summary-level categories to avoid cluttering the plot
top_categories = [
    'Food expenditures',
    'Food purchased from stores',
    'Food purchased from restaurants'
]
df_filtered = df_plot[df_plot['Food expenditures, summary-level categories'].
    ↪ isin(top_categories)]

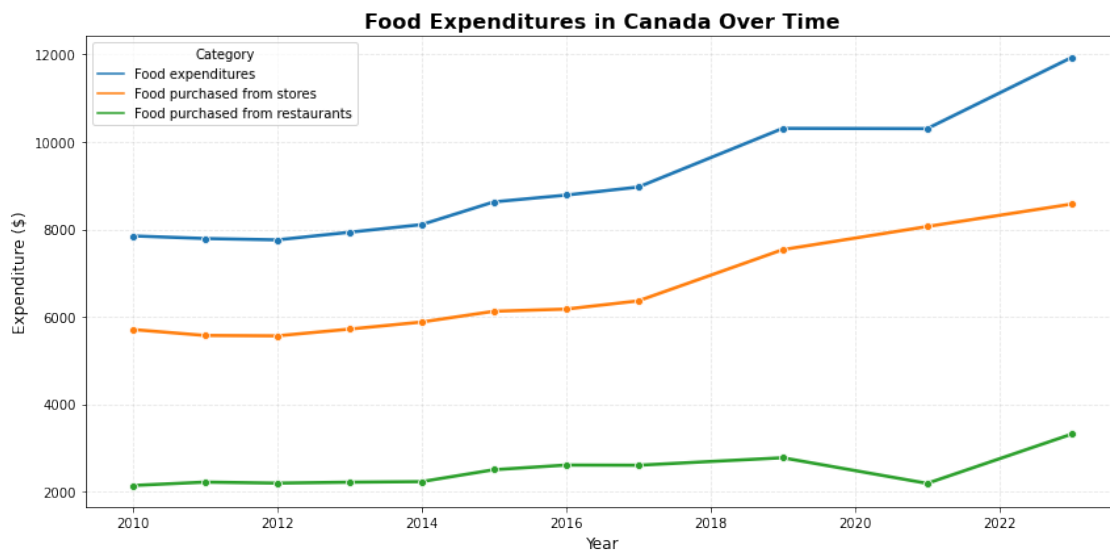
# Initialize the matplotlib figure canvas
plt.figure(figsize=(12,6))

# Create the line plot using seaborn
sns.lineplot(data=df_filtered,
             x='REF_DATE',
             y='VALUE',
             hue='Food expenditures, summary-level categories',
             marker='o',
             linewidth=2.5)

# Customize plot titles, labels, legend, and grid styling
```

```
plt.title("Food Expenditures in Canada Over Time",
          fontsize=16, fontweight='bold')
plt.xlabel("Year", fontsize=12)
plt.ylabel("Expenditure ($)", fontsize=12)
plt.legend(title="Category")
plt.grid(True, linestyle='--', alpha=0.3)
plt.tight_layout()

# Display the final visualization
plt.show()
```



```
[6]: # Filter data for Canada and the latest available year
latest_year = food_spending['REF_DATE'].max()
df_snapshot = food_spending[
    (food_spending['GEO'] == 'Canada') &
    (food_spending['REF_DATE'] == latest_year)
].copy()

# Convert VALUE to numeric and drop missing values
df_snapshot['VALUE'] = pd.to_numeric(df_snapshot['VALUE'], errors='coerce')
df_snapshot = df_snapshot.dropna(subset=['VALUE'])

df_snapshot = df_snapshot[df_snapshot['Food expenditures, summary-level_
categories'].isin(top_categories)]

# Create a clean horizontal bar plot for a single year
plt.figure(figsize=(10, 4))
sns.barplot(
```

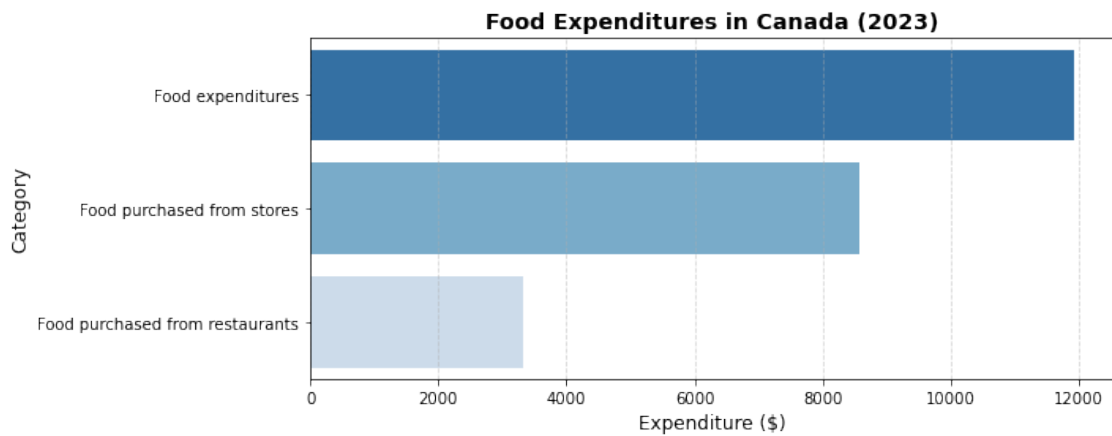
```

data=df_snapshot,
y='Food expenditures, summary-level categories',
x='VALUE',
palette='Blues_r'
)

# Customizations
plt.title(f"Food Expenditures in Canada ({latest_year})", fontsize=14,
fontweight='bold')
plt.xlabel("Expenditure ($) ", fontsize=12)
plt.ylabel("Category", fontsize=12)
plt.grid(axis='x', linestyle='--', alpha=0.5)
plt.tight_layout()

plt.show()

```



Data Source / Citation: > Statistics Canada. Table 11-10-0125-01 Detailed food spending, Canada, regions and provinces
> DOI: <https://doi.org/10.25318/1110012501-eng>